

NOV 4

Varimax criteria

$\hat{L} \rightarrow p \times m$ (sample factor estimation)

$T \rightarrow m \times m$ (rotation matrix - orthogonal matrix)

$\uparrow$ currently unknown

Take $\hat{L}^* = \hat{L} T$ $[\hat{l}_{ij}^*] = \hat{L}^*$ ← rotated matrix L

Define $\tilde{l}_{ij} = \frac{\hat{l}_{ij}^*}{\hat{h}_i}$ $\left((\hat{h}_i^*)^2 = \sum_{j=1}^m (\hat{l}_{ij}^*)^2 \right)$
normalization $\rightarrow$ $1 \leq i \leq p, 1 \leq j \leq m$ $\uparrow$ communality

$$x_i = \mu_i + l_{i1}^* F_1^* + l_{i2}^* F_2^* + \dots + l_{im}^* F_m^* + \varepsilon_i$$

$$V_j = \text{Var}(j^{\text{th}} \text{ col of } M) \quad 1 \leq j \leq m$$

$$= E(X^2) - [E(X)]^2$$

$$= E\left(\frac{\sum \hat{l}_{ij}^4}{p}\right) - \left(\frac{\sum \hat{l}_{ij}^2}{p}\right)^2$$

$$M = [\hat{l}_{ij}^2]$$

scaled rotated matrix
 $\hat{L}$ normalization
 $\hat{L}^*$ rotated matrix

$$= g(t_{ij}) \quad \leftarrow \text{some expression of } t_{ij}$$

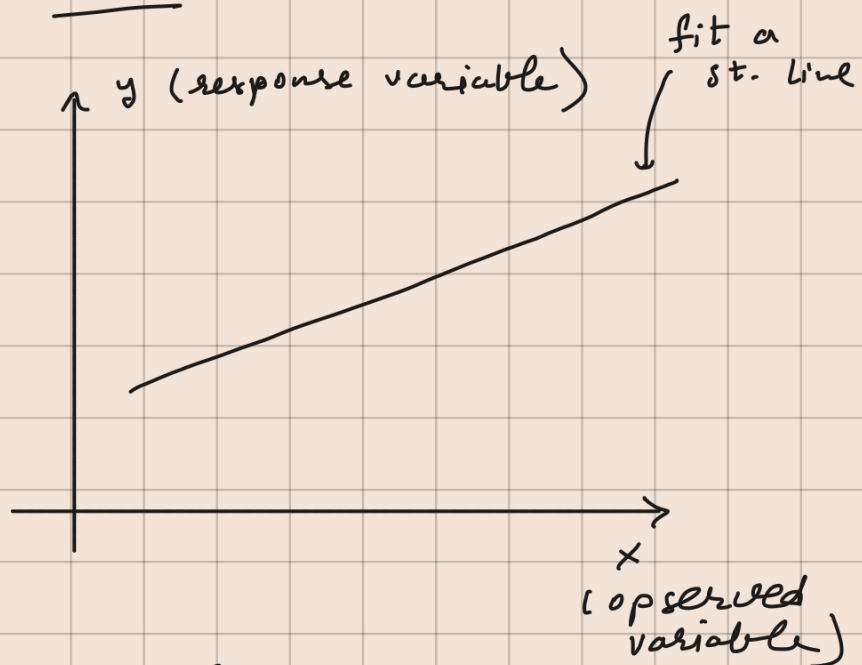
Total spread or var in M (wrt m -factors)

$$= \sum_{j=1}^m V_j$$

We try to maximize this

Regression

in 2D



observed var: $z_1, \dots, z_r$ ($r \geq 1$)

response var $\rightarrow y$

$$y = \beta_0 + \beta_1 z_1 + \dots + \beta_r z_r + \varepsilon$$

ε white noise

$$E(\varepsilon) = 0$$

$$V(\varepsilon) = \sigma^2$$

say we have r -features and n -samples (independent)

$$y_1 = \beta_0 + \beta_1 z_{11} + \dots + \beta_r z_{1r} + \varepsilon_1$$

$\vdots$

$$y_n = \beta_0 + \beta_1 z_{n1} + \dots + \beta_r z_{nr} + \varepsilon_n$$

$$\boxed{Y = Z\beta + \varepsilon}$$

$n \times 1$

$$Z = [z_{ij}] \quad \begin{matrix} i=1 \dots n \\ j=1 \dots, p+1 \end{matrix}$$

$$z_0 = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}_{n \times 1}$$

$$Z = \begin{pmatrix} 1 & z_{11} & \dots & z_{1p} \\ \vdots & \vdots & & \vdots \\ z_{n1} & \dots & \dots & z_{np} \end{pmatrix}$$

$$\text{error} = \Sigma = Y - Z\beta$$

$$S(\beta) = \Sigma^T \Sigma = (Y - Z\beta)^T (Y - Z\beta)$$

least sq method:

$$\text{minimize } \Sigma^T \Sigma = \|\Sigma\|^2$$

$$\text{i.e. } \min_{\beta} S(\beta)$$

$$S(\beta) = Y^T Y + \beta^T Z^T Z \beta - \beta^T Z^T Y - Y^T Z \beta$$

$\uparrow$
 1×1 matrix

$$= Y^T Y + \beta^T Z^T Z \beta - 2 \beta^T Z^T Y$$

$$\frac{\partial S}{\partial \beta} = 0$$

$$2 z^T z \beta - 2 z^T y = 0$$

$$z^T z \beta = z^T y$$

if $z^T z$ is invertible i.e. $\text{rank}(z^T z) = r+1$

$$\text{then } \beta = (z^T z)^{-1} z^T y$$

$$\frac{d^2 S}{d \beta^2} = 2 z^T z$$

↑ positive
semi definite

∴ the problem is convex
⇒ stationary pt is
optimal solⁿ

$\hat{\beta}$ is an estimated value of β

$$\begin{aligned} S(\beta) &= (y - z\beta)^T (y - z\beta) \\ &= (y - z\beta + z\hat{\beta} - z\hat{\beta})^T \end{aligned}$$

$$* (y - z\beta + z\hat{\beta} - z\hat{\beta})$$

$$= \left((y - z\hat{\beta}) + z(\hat{\beta} - \beta) \right)^T \left((y - z\hat{\beta}) + z(\hat{\beta} - \beta) \right)$$

$$= (y - z\hat{\beta})^T (y - z\hat{\beta})$$

$$+ (\hat{\beta} - \beta)^T z^T z (\hat{\beta} - \beta)$$

$$+ 2(y - z\hat{\beta})^T z (\hat{\beta} - \beta)$$

$$\rightarrow \|(\hat{\beta} - \beta)z\|^2$$

$$(y - z\hat{\beta})z$$

$$\hat{\beta} = (z^T z)^{-1} z^T y$$

$$= (y - z(z^T z)^{-1} z^T y)^T z$$

historical
y

$$= 0$$

$$S(\hat{\beta}) = (y - z\hat{\beta})^T (y - z\hat{\beta}) + \|(\hat{\beta} - \beta)z\|^2$$

Nov 6

$$y = \beta_0 + \beta_1 z_1 + \dots + \beta_r z_r + \varepsilon$$

$$\equiv y = \beta_0 z_0 + \beta_1 z_1 + \dots + \beta_r z_r + \varepsilon \quad [z_0 = 1]$$

$$\begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} \beta_0 \cdot 1 \\ \vdots \\ \beta_0 \cdot 1 \end{pmatrix} + \begin{pmatrix} z \\ \text{known} \end{pmatrix} \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_r \end{pmatrix} + \begin{pmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{pmatrix}$$

$n = \text{no. of sample data}$

unknown but deterministic

$$y = Z\beta + \varepsilon$$

$n \times 1$ $n \times (r+1)$ $(r+1) \times 1$ $n \times 1$

$$E(\varepsilon) = 0$$

$$V(\varepsilon) = \sigma^2 I$$

error in i^{th} and j^{th} observation is independent

minimize $\varepsilon^T \varepsilon$

$$\equiv \text{minimize } S(\beta) = (y - Z\beta)^T (y - Z\beta)$$

for sample part

$$\hat{\beta} = (Z^T Z)^{-1} Z^T y$$

this is an estimator for β by sampling process

assuming $Z^T Z$ is invertible

$$y = z\beta + \varepsilon \rightarrow \text{observed value } \int y$$

$$\hat{y} = z\hat{\beta} \rightarrow \text{predicted}$$

Define $H = z(z^T z)^{-1} z^T$
 $(n \times n)$

H : Hat matrix

$$I - H = I - z(z^T z)^{-1} z^T \leftarrow \begin{array}{l} \text{symmetric} \\ \text{idempotent matrix} \end{array}$$

$$(I - H)^2 = I - H$$

①

$$z^T \hat{\varepsilon} = 0$$

this is orthogonality

$$\therefore z^T \hat{\varepsilon} = z^T (I - H)y = 0$$

$$\hat{\varepsilon} = y - \hat{y}$$

$$= y - z\hat{\beta}$$

$$= y - z((z^T z)^{-1} z^T y)$$

$$= (I - H)y$$

$$(I - H)z = z - Hz$$

$$= z - z(z^T z)^{-1} z^T z$$

$$= z - z = 0$$

$$\Rightarrow z^T (I - H)^T = 0$$

$$\Rightarrow z^T (I - H) = 0$$

②

$$z^T \hat{\varepsilon} = 0$$

$$\begin{bmatrix} 1 & \dots & \dots & 1 \end{bmatrix}_{(n+1) \times n}$$

$$\begin{bmatrix} \hat{\varepsilon}_1 \\ \vdots \\ \hat{\varepsilon}_n \end{bmatrix}_{n \times 1}$$

$$= 0$$

$$\Rightarrow \boxed{\hat{\varepsilon}_1 + \dots + \hat{\varepsilon}_n = 0}$$

$$\mathbf{1}^T \hat{\varepsilon} = 0$$

$$\Rightarrow \mathbf{1}^T (\mathbf{y} - \mathbf{y}^T) = 0$$

$$\Rightarrow \mathbf{1}^T \mathbf{y} = \mathbf{1}^T \hat{\mathbf{y}}$$

$$\Rightarrow \frac{\sum_{i=1}^n y_i}{n} = \frac{\sum_{i=1}^n \hat{y}_i}{n}$$

$\Rightarrow$ observed mean = predicted response mean

$$(3) \quad S(\hat{\beta}) = \hat{\varepsilon}^T \hat{\varepsilon} \\ = (\mathbf{I} - \mathbf{H}) \mathbf{y}^T (\mathbf{I} - \mathbf{H}) \mathbf{y}$$

$$= \mathbf{y}^T (\mathbf{I} - \mathbf{H})^T (\mathbf{I} - \mathbf{H}) \mathbf{y}$$

$$(\mathbf{I} - \mathbf{H})^2 = \mathbf{I} - \mathbf{H}$$

$$= \mathbf{y}^T (\mathbf{I} - \mathbf{H}) \mathbf{y}$$

$$= \mathbf{y}^T \mathbf{y} - \mathbf{y}^T \mathbf{H} \mathbf{y}$$

$$\Rightarrow \mathbf{y}^T \mathbf{y} = \mathbf{y}^T \mathbf{H} \mathbf{y} + \hat{\varepsilon}^T \hat{\varepsilon}$$

$$\left[\mathbf{H} \cdot \mathbf{y} = \underbrace{\mathbf{z} (\mathbf{z}^T \mathbf{z}^{-1}) \mathbf{z}^T \mathbf{y}}_{\hat{\beta}} = \mathbf{z} \hat{\beta} = \hat{\mathbf{y}} \right]$$

$$= \mathbf{y}^T \mathbf{H}^T \mathbf{H} \mathbf{y} \\ = \hat{\mathbf{y}}^T \hat{\mathbf{y}}$$

and also $\mathbf{H}^2 = \mathbf{H}$

symmetric, idempotent

$$\mathbf{y}^T \mathbf{y} = \hat{\mathbf{y}}^T \hat{\mathbf{y}} + \hat{\varepsilon}^T \hat{\varepsilon}$$

$$\mathbf{y}^T \mathbf{y} - n \bar{y}^2 = \hat{\mathbf{y}}^T \hat{\mathbf{y}} - n \bar{\hat{\mathbf{y}}}^2 + \hat{\varepsilon}^T \hat{\varepsilon}$$

$$\text{as } \bar{y} = \bar{\hat{\mathbf{y}}}$$

$$\Rightarrow \sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n (\hat{y}_i - \bar{\hat{y}})^2 + \sum_{i=1}^n (\hat{\epsilon}_i - 0)^2$$

$\Rightarrow$ Total sum of sq (TSS) of variation in observed value from mean = RSS + ESS
 (Total sum of sq of regressed values from their mean) (error sum of sq)

$$TSS = RSS + ESS$$

for good fit ESS is min

$$TSS \geq RSS$$

$$R^2 \text{ value} = 1 - \frac{ESS}{TSS} = \frac{RSS}{TSS}$$

$$R^2 \in [0, 1]$$

$$\text{if } R^2 \rightarrow 1 \Rightarrow ESS \rightarrow 0$$

Best fit

$$\text{if } R^2 \rightarrow 0 \Rightarrow ESS \sim TSS$$

$$\Rightarrow \hat{y}_i = \bar{y}$$

then $\hat{\mathbf{y}} = \begin{pmatrix} \bar{y} \\ \vdots \\ \bar{y} \end{pmatrix}$

$$\hat{y} = z \hat{\beta} = \bar{y} \cdot 1$$

$$z \begin{pmatrix} \hat{\beta}_0 \\ \vdots \\ \hat{\beta}_r \end{pmatrix} = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}$$

with $z \rightarrow 1^{\text{st}}$ col is 1

worst possible regression model, $R^2=0$

$$(\hat{\beta}_0 - \bar{y}) \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} + \hat{\beta}_1 \begin{pmatrix} z_{11} \\ \vdots \\ z_{n1} \end{pmatrix} + \dots + \hat{\beta}_r \begin{pmatrix} z_{1r} \\ \vdots \\ z_{nr} \end{pmatrix} = 0$$

as z has full rank $\Rightarrow$ col of z are linearly independent

linear combination of cols of z

$\Rightarrow$ each coef = 0

$$\Rightarrow \hat{\beta}_0 = \bar{y} \quad \text{and} \quad \hat{\beta}_j = 0 \quad \forall 1 \leq j \leq r$$

But this is NOT regression.

increasing $R^2 \rightarrow$ reducing ESS
this can be done by more observed features
but can cause overfitting

Adding new features will never increase ESS

say we add feature z , it may set $\beta_z = 0$ to maintain same ESS

So we can Hack R^2 by introducing new random features and overfitting the model

$$\text{Adjusted } R^2 = 1 - \frac{\text{mean sum of sq}}{\text{Total mean sum of sq}}$$

$$= 1 - \frac{\frac{ESS}{n - (r+1)}}{\left(\frac{TSS}{n-1} \right)}$$

$$R_a^2 = 1 - (1 - R^2) \left(\frac{n-1}{n - (r+1)} \right)$$

as $r \uparrow$

$R^2 \uparrow$

$1 - R^2 \downarrow$

$\frac{1}{n - r + 1} \uparrow$

R_a^2 can $\uparrow$

7 Nov

We are giving geometric interpretation of OLS via eigenvectors of $Z^T \cdot Z$

$$z^T z \rightarrow (r+1)(r+1)$$

↳ full rank

$\hookrightarrow$ sym, rel def

$$z^T z = \sum_{i=1}^{z+1} \lambda_i e_i e_i^T \quad (\text{spectral decomposition})$$

$$(Z^T Z)^{-1} = \sum_{i=1}^{n+1} \frac{1}{\lambda_i} e_i e_i^T$$

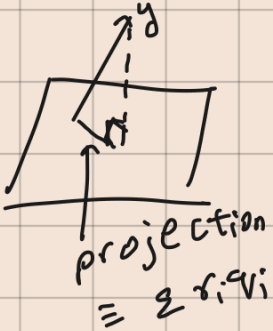
$\begin{matrix} & & \uparrow & & \uparrow \\ & & (i+1) \times 1 & & 1 \times (i+1) \end{matrix}$

$$M = Z(Z^T Z)^{-1} Z^T = \sum_{i=1}^{r+1} \lambda_i^{-1} z_i z_i^T z_i^T$$

$$= \sum_{i=1}^{s+1} \underset{n \times 1}{q_i} \underset{1 \times n}{q_i^T}, \quad \boxed{q_i = \lambda^{-1/2} z e_i}$$

q_i are in R_n and orthonormal and span the column space of Z

If we consider a space spanned by $\{v_1, \dots, v_{r+1}\}$ in $\mathbb{R}^n$ and we take a projection of vector $y \in \mathbb{R}^n$ on this space



$$\sum_{i=1}^{s+1} (q_i^T y) q_i = \sum_{i=1}^{s+1} (q_i q_i^T) y = Hy = z\hat{\beta} = \hat{y}$$

Result :- (a) $E(\hat{\beta}) = \beta \Rightarrow \hat{\beta}$ is unbiased estimator of β

(b) $E(\hat{\beta}) = E(y - \hat{y}) = 0$
and $\text{Cov}(\hat{\beta}) = \sigma^2 (I - H)$

where $\text{cov}(\mathbf{z}_i) = \sigma^2 \mathbf{I}$

$$(c) \quad E(\hat{\varepsilon}^T \hat{\varepsilon}) = (n-r-1) \sigma^2$$

$$\Rightarrow E \left(\frac{\hat{\varepsilon}^T \hat{\varepsilon}}{n-r-1} \right) = \sigma^2$$

↑ unbiased estimator of σ^2

$$(d) \quad \hat{\beta} + \hat{\varepsilon} \text{ are independent or } \text{cov}(\hat{\beta}, \hat{\varepsilon}) = 0$$

proofs

$$(a) \quad E(\hat{\beta}) = E((Z^T Z)^{-1} Z^T Y)$$

$$\uparrow \\ (Z\beta + \varepsilon)$$

$$= E \left(\beta + \text{const. matrix} \cdot \varepsilon \right)$$

$$\uparrow \quad E(\varepsilon) = 0$$

$$= E(\beta)$$

$$(b) \quad E(\hat{\varepsilon}) = E((I-H)Y)$$

$$= E((I-H)(Z\beta + \varepsilon))$$

$$= E \left[\underbrace{(I-H)Z\beta}_{\uparrow} + (I-H)\underbrace{\varepsilon}_{\uparrow} \right] = 0$$

0 (last class)

0

$$\begin{aligned} \text{cov}(\hat{\varepsilon}) &= E \left[(\hat{\varepsilon} - E(\hat{\varepsilon})) (\hat{\varepsilon} - E(\hat{\varepsilon}))^T \right] \\ &= E(\hat{\varepsilon} \hat{\varepsilon}^T) \end{aligned}$$

$\uparrow \quad 0 \qquad \qquad \qquad \uparrow \quad 0$

$$= E \left[(I-H) Y Y^T (I-H)^T \right]$$

$$= (I-H) E(Y Y^T) (I-H)$$

$$= (I-H) E \left[(Z\beta + \varepsilon) (Z\beta + \varepsilon)^T \right] (I-H)$$

$$= (I-H) \times E \left[\underset{\substack{\uparrow \\ \text{constant}}}{Z\beta\beta^T Z + Z\beta\varepsilon^T + \varepsilon\beta^T Z^T + \varepsilon\varepsilon^T} \right] \times (I-H)$$

$$= (I-H) E(\varepsilon\varepsilon^T) (I-H) \quad \text{using } (I-H)Z=0$$

$$= (I-H) \text{cov}(\varepsilon) (I-H) \quad \text{and } E(\varepsilon)=0$$

$$= (I-H) \sigma^2 I (I-H)$$

$$= \sigma^2 (I-H)^2 = \sigma^2 (I-H)$$

$$\textcircled{c} E(\hat{\varepsilon}^T \varepsilon) = E \left[((I-H) Y)^T (I-H) Y \right]$$

$$= E \left[Y^T \underbrace{(I-H)^T (I-H)}_{= I-H} Y \right]$$

$$= (I-H)(I-H)$$

$$= I-H$$

$$= E \left[Y^T (I-H) Y \right]$$

$$= E \left[(Z\beta + \varepsilon)^T (I-H) (Z\beta + \varepsilon) \right]$$

$$= E \left[\varepsilon^T (I-H) \varepsilon \right]$$

$$\text{as } (I-H)Z=0$$

$$= E \left[(\varepsilon_1 \dots \varepsilon_n) \begin{pmatrix} 1-h_{11} & -h_{12} & \dots & -h_{1n} \\ -h_{21} & 1-h_{22} & \dots & -h_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ -h_{n1} & -h_{n2} & \dots & 1-h_{nn} \end{pmatrix} \begin{pmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{pmatrix} \right]$$

$$= E \left[\sum_{i=1}^n (1 - h_{ii}) \varepsilon_i^2 + \underbrace{\sum_{\substack{i,j \\ i \neq j}} 2 h_{ij} \varepsilon_i \varepsilon_j} \right]$$

$$\text{cov}(\varepsilon) = \sigma^2 I$$

$$\text{cov}(\varepsilon_i, \varepsilon_j) = 0$$

$$E(\varepsilon_i, \varepsilon_j) = 0$$

$$\text{trace}(I - H) \sigma^2$$

$$= \text{trace}(I - H) \sigma^2 = \sigma^2 [n - (r+1)]$$

$\rightarrow n \times n$
 H : idempotent
 $H^2 = H$

0, 1 are only eigenvalues of $H_{n \times n}$
 and $\text{rank } H = r+1$

$\left. \begin{array}{l} 0 \rightarrow n - (r+1) \text{ times} \\ 1 \rightarrow r+1 \text{ times} \end{array} \right\} \rightarrow \text{trace of RRE}$

as it is transformation of $z^T z$

$$\textcircled{d} \text{Cov}(\hat{\beta}, \hat{\varepsilon}) = 0$$

$$\text{Cov}(\hat{\beta}, \hat{\varepsilon}) = E \left[\left(\hat{\beta} - E(\hat{\beta}) \right) \left(\hat{\varepsilon} - E(\hat{\varepsilon}) \right)^T \right]$$

$\uparrow \quad \quad \quad \uparrow$
 $\beta \quad \quad \quad 0$

$$= E \left(\left(\hat{\beta} - \beta \right) \hat{\varepsilon}^T \right)$$

$\uparrow$
 $(z^T z)^{-1} z^T y$
 $\uparrow$
 $z\beta + \varepsilon$

$$= E \left[(z^T z)^{-1} z^T \varepsilon \hat{\varepsilon}^T \right]$$

$$= E \left[(z^T z)^{-1} z^T \varepsilon y^T \underbrace{(I - H)^T}_{(z\beta + \varepsilon)^T (I - H)} \right]$$

$$= (z^T z)^{-1} z^T \underbrace{E(\varepsilon \varepsilon^T)}_{\substack{\uparrow \\ \text{Cov}(\varepsilon) \\ = \sigma^2 I}} (I - H)$$

$$= \sigma^2 (z^T z)^{-1} \underbrace{z^T (I - H)}_0$$

$$= 0$$

$$\begin{aligned}
 (e) \quad \boxed{\text{Cov}(\hat{\beta})} &= E \left[\left(\hat{\beta} - E(\hat{\beta}) \right) \left(\hat{\beta} - E(\hat{\beta}) \right)^T \right] \\
 &= E \left[\left(\hat{\beta} - \beta \right) \left(\hat{\beta} - \beta \right)^T \right] \\
 &= E \left[\left((z^T z)^{-1} z^T y \right) \left((z^T z)^{-1} z^T z \beta \right)^T \right] \\
 &= E \left[\left((z^T z)^{-1} z^T (y - z \beta) \right) \left((z^T z)^{-1} z^T (y - z \beta) \right)^T \right] \\
 &= (z^T z)^{-1} z^T E(\varepsilon \varepsilon^T) z (z^T z)^{-1} \\
 &= \sigma^2 (z^T z)^{-1} z^T z (z^T z)^{-1} \\
 &= \boxed{\sigma^2 (z^T z)^{-1}}
 \end{aligned}$$

11 Nov 2023

classmate

Date

Page

Model of $Y = Z\beta + \epsilon$ $\text{rank}(Z) = n+1$

$$E(\epsilon) = 0$$

$$\text{Var}(\epsilon) = \sigma^2 I$$

$$\hat{\beta} = (Z^T Z)^{-1} Z^T y \rightarrow \text{Ordinary Least Square (OLS)}$$

$$\hat{y} = Z\hat{\beta}$$

$$\hat{e} = y - \hat{y}$$

$$H_{\text{non}} = Z(Z^T Z)^{-1} Z^T$$

BLUE

Best Linear Unbiased Estimator

$$E(S) = \theta \quad \text{Unbiased.}$$

linear state of sample observation

Among all multiple choices for S i.e. (which is unbiased as well as some linear function to sample) the one with min-
variance.

That statistics is called "BLUE Test".

Gauss Markov Theorem

- Among all possible linear unbiased estimator of β vector, the one obtained from OLS, i.e. $\hat{\beta}$ is BLUE.

Pr:

We have already seen that the OLS estimator of β is

$$\hat{\beta} = (Z^T Z)^{-1} Z^T y$$

(2×1) $(n+1) \times n$ $(n+1) \times 1$ $(n+1) \times n$

$(n+1) \times n$ constant matrix
 becz Z is fixed matrix known to us.

$$\therefore \hat{\beta} = A y$$

linear function of y

 $\Rightarrow$ Linear

We have already proved that $E(\hat{\beta}) = \beta$.

 $\Rightarrow$ Unbiased estimator

Now it is to prove that it has least Variance

$$\text{Cov}(\hat{\beta}) = \sigma^2 (Z^T Z)^{-1}$$

(Done in last class)

Let $\tilde{\beta}$ be any other unbiased linear estimator for β , that is,

$$\left. \begin{aligned} \tilde{\beta} &= A y \\ E(\tilde{\beta}) &= \beta \end{aligned} \right\}$$

for some constant $(n+1) \times n$ matrix A

$$\begin{aligned} \tilde{\beta} &= A y \\ &= A(\beta + e) \\ &= A Z \beta + A e \end{aligned}$$

Also

$$E(\tilde{\beta}) = \beta$$

$$\Rightarrow E(Ay) = \beta$$

$$\Rightarrow E(Az\beta + Ae) = \beta$$

$$\Rightarrow Az\beta E(1) + AE(e) = \beta$$

$$\Rightarrow \underbrace{Az\beta}_{(n+1) \times n \quad n \times (n+1)} = \beta$$

AZ is $(n+1) \times (n+1) \rightarrow$ Identity matrix.

$$\tilde{\beta} = \beta + AE^{-1}S^T y - A\beta = \hat{\beta}$$

Also

$$\hat{\beta} = (Z^T Z)^{-1} Z^T y$$

$$= (Z^T Z)^{-1} Z^T (Z\beta + e)$$

$$= Z^T Z^{-1} (Z^T Z \beta) + (Z^T Z)^{-1} Z^T e$$

$$= \beta + (Z^T Z)^{-1} Z^T e$$

$$\tilde{\beta} - \hat{\beta} = \underbrace{\left(A - (Z^T Z)^{-1} Z^T \right)}_{\substack{(n+1) \times n \\ \text{constant Matrix}}} e$$

$n \times 1$ random

Claim:-

$$\text{Cov}(\tilde{\beta} - \hat{\beta}, \hat{\beta}) = 0 \quad (*)$$

$$\text{Var}(X) = E[(X - E(X))^T (X - E(X))]$$

$$\Rightarrow \text{Var}(\tilde{\beta}) = \text{Var}(\hat{\beta} + (\tilde{\beta} - \hat{\beta}))$$

$$= \text{Var}(\hat{\beta}) + \text{Var}(\tilde{\beta} - \hat{\beta}) + 2 \text{Cov}(\hat{\beta}, \tilde{\beta} - \hat{\beta})$$

(As per claim)

$$\Rightarrow \text{Var}(\tilde{\beta}) = \text{Var}(\hat{\beta}) + \text{Var}(\tilde{\beta} - \hat{\beta})$$

$$\Rightarrow \text{Var}(\tilde{\beta}) \geq \text{Var}(\hat{\beta})$$

At linear unbiased estimator $\tilde{\beta}$ of β

$\Rightarrow \hat{\beta}$ has least variance

$\Rightarrow \hat{\beta}$ is BLUE of β .

Back to
Claim:

$$\text{Cov}(\tilde{\beta} - \hat{\beta}, \hat{\beta})$$

$$= \text{Cov}(\hat{\beta}, \tilde{\beta} - \hat{\beta})$$

$$= \text{Cov}(CE, DE)$$

$$\tilde{\beta} - \hat{\beta} = (A - (Z^T Z^{-1}) Z^T) E$$

$$= DE$$

$$\hat{\beta} = \beta + (Z^T Z)^{-1} Z^T E$$

Shifting const

CE

so we can ignore it
while cov calculate

Shifting by constant factor

does not account in variance
(Because there is no mobility)

Shifting by a constant vector. (no randomness)
does not alter variance or covariance.

$$= \text{Cov}(\hat{\beta}, \tilde{\beta} - \hat{\beta})$$

$$= \text{Cov}(CE, DE)$$

$$= C \text{Cov}(E, E) D^T$$

$$= C \sigma^2 I D^T$$

$$= \sigma^2 C D^T$$

$$= \sigma^2 (Z^T Z^{-1}) Z^T (A - (Z^T Z)^{-1} Z^T)^T$$

$$= \sigma^2 (Z^T Z^{-1}) Z^T (A^T - \frac{Z(Z^T Z^{-1})^T}{Z(Z^T Z^{-1})^T})$$

$$= \sigma^2 (Z^T Z)^{-1} \left[Z^T A^T - Z^T Z (Z^T Z)^{-1} Z^T \right]$$

$$= \sigma^2 (Z^T Z)^{-1} \left[\begin{matrix} (AZ)^T - I \\ I \end{matrix} \right]$$

$$= 0 \quad \text{since } AZ = I_{n+1}$$

MLE of β & σ^2 ?

$$\text{Cov}(e) = \sigma^2 I$$

$$E(\hat{\beta}) = \beta$$

$$\text{Cov}(\hat{\beta}) = \sigma^2 (Z^T Z)^{-1}$$

$$\text{Assume } e \sim N_n(0, \sigma^2 I)$$

$$\Leftrightarrow e_i \sim N(0, \sigma^2), \quad \forall i=1, 2, \dots, n$$

e_i 's are independent w.v.

$$\text{Also, } Y = Z\beta + e$$

$$E(Y) = Z\beta, \quad \text{Var}(Y) = \text{Var}(e) = \sigma^2 I$$

$$\Rightarrow Y \sim N_n(Z\beta, \sigma^2 I) \Rightarrow Y_i\text{'s are independent w.v.}$$

Claim:

$$\hat{\beta}_{MLE} = \hat{\beta}$$

$$\hat{\beta} = (Z^T Z)^{-1} Z^T Y \quad \text{linear in } Y$$

$$\hat{\beta} \sim N_{n+1}(\beta, \sigma^2 (Z^T Z)^{-1})$$

Claim

$$\hat{\beta}_{MLE} = \hat{\beta}$$

$$\sigma^2_{MLE} = \frac{1}{n} \hat{\epsilon}^T \hat{\epsilon}$$

$$\hat{\epsilon} = y - \hat{y} = y - Z\hat{\beta}$$

$$= (I - H)\epsilon$$

Given a sample data: $(z_1, z_2, \dots, z_n, y)$

Proof:

n samples: $y = Z\beta + \epsilon$
randomJoint pdf of ϵ vector

$$f(\epsilon) = \frac{1}{(2\pi)^{n/2} |\sigma^2 I|^{n/2}} \exp\left(-\frac{1}{2} (y - Z\beta)^T (\sigma^2 I)^{-1} (y - Z\beta)\right)$$

$$\begin{aligned} &= \text{trace}(AB) \\ &= \text{trace}(BA) \end{aligned}$$

$$\ln L(\beta, \sigma^2) = \ell(\beta, \sigma^2)$$

$$= -\frac{n}{2} \ln |\sigma^2 I| - \frac{1}{2} \text{trace}((\sigma^2 I)^{-1} (y - Z\beta)(y - Z\beta)^T) + \text{constant}$$

concave func.
stationary pt
is local maximum

(factor of 2n)

$$\max L \equiv \max \ell(\beta, \sigma^2)$$

$$\frac{\partial \ell}{\partial \beta} = 0 \Rightarrow \hat{\beta}_{MLE} \text{ is a sol.}$$

$$\hat{\beta}_{MLE} = \hat{\beta}$$

Cramer

$$\Rightarrow Z^T (y - Z\hat{\beta}) (\sigma^2 I)^{-1} = 0$$

$$\Rightarrow Z^T (y - Z\hat{\beta}) \Rightarrow Z^T y = Z^T Z \hat{\beta}$$

$$\Rightarrow \hat{\beta} = (Z^T Z)^{-1} (Z^T y) = \hat{\beta}$$

12-Nov.

MLE:

est. $\epsilon \sim N_n(0, \sigma^2 I)$

$\epsilon_i' \sim N(0, \sigma^2), \forall i$

ϵ_i 's are independent.

$Y = Z\beta + \epsilon$

$Y \sim N_n(Z\beta, \sigma^2 I)$

y_i 's are independent (bec. there covariance 0)

$$\frac{L(\beta, \sigma^2)}{L(\beta, \sigma^2)} = \frac{1}{(2\pi)^{n/2} |\Sigma|^{n/2}} \exp \left(-\frac{1}{2} \underbrace{(Y - Z\beta)^T \Sigma^{-1} (Y - Z\beta)}_{1 \times 1 \text{ trace}(\dots)} \right)$$

= " $\exp \left(-\frac{1}{2} \text{trace}(\Sigma^{-1} (Y - Z\beta)(Y - Z\beta)^T) \right)$

$l(\beta, \sigma^2) = \ln L(\beta, \sigma^2) = -\frac{n}{2} \ln |\Sigma| - \frac{1}{2} \text{trace}(\Sigma^{-1} (Y - Z\beta)(Y - Z\beta)^T) + \text{constant.}$

↳ concave f^n .

$\max l(\beta, \sigma^2) \equiv \min (Y - Z\beta)(Y - Z\beta)^T$
↳ convex.

$\Rightarrow \boxed{\beta_{MLE} = (Z^T Z)^{-1} Z^T Y = \hat{\beta}}$

$$\begin{aligned} l(\hat{\beta}, \sigma^2) &= -\frac{n}{2} \ln |\Sigma| - \frac{1}{2} \text{trace}(\underbrace{\Sigma^{-1}}_{\hat{\Sigma}^{-1}} (Y - \hat{\gamma})(Y - \hat{\gamma})^T) \\ &= -\frac{n}{2} \ln |\Sigma| - \frac{1}{2} \text{trace}(\underbrace{\hat{\Sigma}^T (\sigma^2 I)^{-1} \hat{\Sigma}}_{1 \times 1}) \\ &= -\frac{n}{2} \ln |\sigma^2 I| - \frac{1}{2} \sigma^2 (\hat{\Sigma}^T \hat{\Sigma}) \end{aligned}$$

$$\hat{\beta} = \beta_{MLE}$$

$$l(\hat{\beta}, \sigma^2) = -\frac{n}{2} \ln(\sigma^2) - \frac{n^2}{2\sigma^2} \left(\frac{\hat{\epsilon}^T \hat{\epsilon}}{n} \right)$$

Model
σ²

$$\sigma_{MLE}^2 = \frac{1}{n} \hat{\epsilon}^T \hat{\epsilon}$$

$$\beta_{MLE} = \hat{\beta}$$

AIM
distribution

Consider

$$\hat{\epsilon} = y - \hat{y} = (I - H)y$$

$$\hat{\epsilon}^T \hat{\epsilon} = y^T (I - H)^T (I - H) y = y^T (I - H) y$$

$$= (Z\beta + \epsilon)^T (I - H) (Z\beta + \epsilon)$$

$$= \epsilon^T (I - H) \epsilon \quad \left[\begin{array}{l} \therefore (I - H)Z = 0 \\ Z^T(I - H) = 0 \end{array} \right]$$

$I - H$ $\rightarrow$ n eigen value which are either 0 or 1
 $n \times n$
matrix

$$\text{rank } H = \text{rank } Z(Z^T Z)^+ Z^T = r + 1$$

$$H^T = H, H^2 = H$$

$\therefore$ Eigen values of H are
0 & 1 only.

$\therefore$ 1 as eigenvalue of H is
repeated $r + 1$ times

$$H \sim \begin{bmatrix} I_{r+1} & 0 \\ 0 & 0 \end{bmatrix}$$

$\Rightarrow I - H$ has rank $n - (r + 1)$ & 0 e.v. of $I - H$ is repeated $r + 1$ times.
 $\downarrow$
all e.v. = 1
 $n - (r + 1)$ times.

we have earlier shown that $E(\hat{\varepsilon}^T \hat{\varepsilon}) = (n-r-1)\sigma^2$

$$\Rightarrow E\left(\frac{\hat{\varepsilon}^T \hat{\varepsilon}}{(n-r-1)}\right) = \sigma^2$$

* $\hat{\beta} \Rightarrow$ MLE & unbiased estimator for β

$\hat{\beta}$ is a linear fn of y

As $\varepsilon \sim \text{Normal}$

$\Rightarrow y \sim \text{Normal}$

$\Rightarrow \hat{\beta} \sim \text{Normal}$

$$\hat{\beta} \sim N_{r+1}(\beta, \sigma^2(Z^T Z)^{-1})$$

$\downarrow$
(r+1) x 1

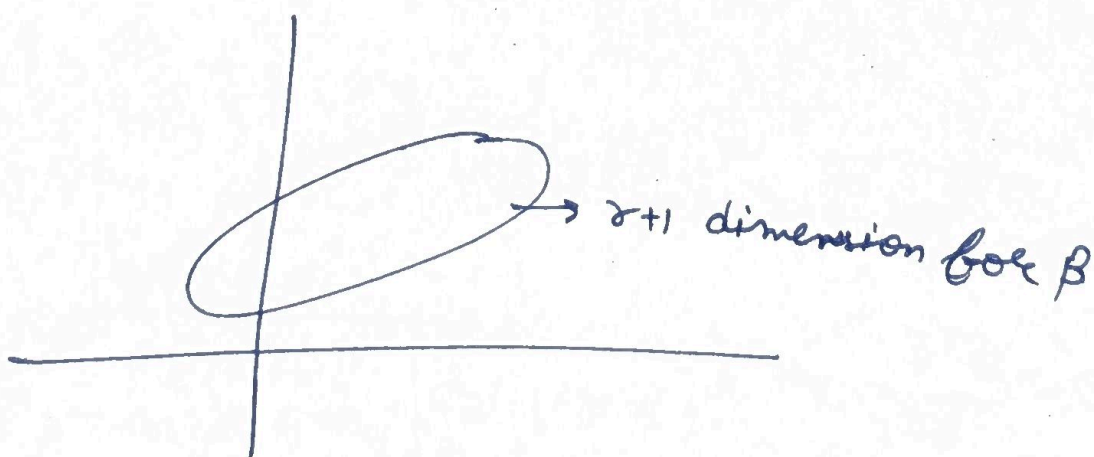
we have proved earlier

$$\text{Cov}(\hat{\beta}) = \sigma^2 (Z^T Z)^{-1}$$

(r+1) x (r+1)

$$E(\hat{\beta}) = \beta$$

$$V = (\hat{\beta} - \beta)^T (\sigma^2 (Z^T Z)^{-1})^{-1} (\hat{\beta} - \beta) \sim \chi^2_{(r+1)}$$



and $V = (n-r-1) \frac{s^2}{\sigma^2} \sim \chi^2_{(n-r-1)}$

$$\Rightarrow \frac{u_i}{\sigma} \sim N(0,1), \forall i$$

$$\Rightarrow \frac{u_i^2}{\sigma^2} \sim \chi^2_{(1)} \quad \forall i$$

Note:-

$$u_i = e_i^T \varepsilon$$

$$\text{cov}(u_i, u_j) = \text{cov}(e_i^T \varepsilon, e_j^T \varepsilon)$$

$$= e_i^T \underbrace{\text{cov}(\varepsilon)} e_j$$

$$= e_i^T (\sigma^2 I) e_j = \sigma^2 \underbrace{e_i^T e_j}_{=0} = 0$$

$\forall i \neq j$

$\therefore e_i$'s are orthonormal

$$\& u_i \sim N(0, \sigma^2), \forall i$$

$\Rightarrow u_i$'s are independent. ($\because u_i$ follow normal random variable distribution)

$$\Rightarrow \sum_{i=1}^{n-r-1} \frac{u_i^2}{\sigma^2} \sim \chi^2_{(n-r-1)}$$

$$\Rightarrow \frac{\sum_{i=1}^{n-r-1} u_i^2}{\sigma^2} \sim \chi^2_{(n-r-1)}$$

$$\Rightarrow \frac{\hat{\varepsilon}^T \hat{\varepsilon}}{\sigma^2} \sim \chi^2_{(n-r-1)} \Rightarrow \boxed{\frac{(n-r-1)s^2}{\sigma^2} \sim \chi^2_{(n-r-1)}}$$

define

$$s^2 = \frac{1}{n-r-1} \hat{\varepsilon}^T \hat{\varepsilon} \rightarrow \text{unbiased estimator of } \sigma^2$$

$$\boxed{\sigma^2_{MLE} = \frac{1}{n} \hat{\varepsilon}^T \hat{\varepsilon}}$$

$$I - H \equiv I - \begin{bmatrix} I_{r+1} & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & I_{n-(r+1)} \end{bmatrix}$$

$$\Rightarrow \text{rank}(I - H) = n - r - 1$$

$$\text{By decomposition of } I - H = \sum_{i=1}^{n-r-1} e_i e_i^T,$$

where e_i 's are eigenvectors corresponding to 1 of $I - H$.

Note:- each e_i , $1 \leq i \leq n-r-1$ is distinct $\therefore I - H$ is symmetric & hence diagonalizable & these e_i 's are orthonormal eigenvectors.

$$\begin{aligned} \text{so, } \hat{\varepsilon}^T \hat{\varepsilon} &= \varepsilon^T (I - H) \varepsilon = \varepsilon^T \sum_{i=1}^{n-r-1} e_i e_i^T \varepsilon \\ &= \sum_{i=1}^{n-r-1} (\underbrace{\varepsilon^T e_i}_{1 \times n}) (\underbrace{e_i^T \varepsilon}_{n \times 1}) \\ &= \sum_{i=1}^{n-r-1} (e_i^T \varepsilon)^2 \\ &= \sum_{i=1}^{n-r-1} u_i^2, \quad u_i = e_i^T \varepsilon, \quad 1 \leq i \leq n-r-1 \end{aligned}$$

$$\text{Let } \varepsilon \in N_n(0, \sigma^2 I)$$

& u_i is a linear combination of ε

$$\text{so } u_i \in N(0, \underbrace{e_i^T (\sigma^2 I) e_i}_{1 \times n \quad n \times n \quad n \times 1}) = N(0, \sigma^2) \because e_i^T e_i = 1 \quad \forall 1 \leq i \leq n-r-1$$

2. $\hat{\beta}, \hat{\varepsilon}$ are independent so U & V are independent.

$$F = \frac{\frac{U}{r+1}}{\frac{V}{n-r-1}} \sim F_{r+1, n-r-1} \text{ (F-distributed)}$$

we had proved earlier

$$\boxed{\text{Cov}(\hat{\beta}, \hat{\varepsilon}) = 0}$$

14-Nov.

14 Nov

$$y = z\beta + \varepsilon$$

$$\text{let } \varepsilon \sim N(0, \sigma^2 I)$$

$$\Rightarrow y \sim N(z\beta, \sigma^2 I)$$

$$\Rightarrow \hat{\beta} \sim N(\beta, \sigma^2 (z^T z)^{-1})$$

$$V = (n-k-1) \frac{s^2}{\sigma^2} \sim \chi^2_{(n-k-1)} \quad s^2 = \frac{\hat{\varepsilon}^T \hat{\varepsilon}}{n-k-1}$$

$$U = (\hat{\beta} - \beta)^T (\sigma^2 (z^T z)^{-1})^{-1} (\hat{\beta} - \beta) \sim \chi^2_{k+1} \quad \rightarrow \text{unbiased estimator of } \sigma^2$$

↑
ellipsoid in $k+1$

$$\text{As } \text{Cov}(\hat{\beta}, \hat{\varepsilon}) = 0$$

$\hat{\beta}$ and $\hat{\varepsilon}$ are indep rand. vec.

U and V are indep rand variable

$$F = \frac{\frac{U}{k+1}}{\frac{V}{n-k-1}} \sim F_{k+1, n-k-1}$$

$$\Rightarrow \frac{(\hat{\beta} - \beta)^T \frac{1}{\sigma^2} (z^T z) (\hat{\beta} - \beta)}{k+1} \times \frac{n-k-1}{(n-k-1) \frac{s^2}{\sigma^2}} \sim F$$

$$\frac{(\hat{\beta} - \beta)^T (z^T z) (\hat{\beta} - \beta)}{(k+1) s^2} \sim F_{k+1, n-k-1}$$

↑ gives ellipse confidence interval for β

Null hyp

$$H_0 : \beta = \beta^*$$

$$H_1 : \beta \neq \beta^*$$

↓
(r+1) × 1

↙ expected / calculated value of β

$$\frac{(\hat{\beta} - \beta^*)^T (Z^T Z) (\hat{\beta} - \beta^*)}{s^2 (r+1)} > F_{r+1, n-r-1}(\alpha)$$

then reject

else fail to reject

CI for $a^T \beta$:

let $a \in \mathbb{R}^{r+1}$

Critical value c

$$= a^T \hat{\beta} \pm \sqrt{(r+1) F_{r+1, n-r-1} s^2 (a^T (Z^T Z)^{-1} a)}$$

for β_j : $a = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \leftarrow j\text{th}$

Ex

$n=8 \quad r=2$
 z_1, z_2, y

data given

model : $y = \beta_0 + \beta_1 z_1 + \beta_2 z_2 + \varepsilon$

compute $\hat{\beta} = (Z^T Z)^{-1} Z^T y = \begin{bmatrix} 2.234 \\ -0.069 \\ 0.291 \end{bmatrix}$

$\hat{y} = \overset{\text{given}}{Z} \hat{\beta}$

$\hat{\varepsilon} = y - \hat{y}$

$R^2 = \frac{\hat{\varepsilon}^T \hat{\varepsilon}}{n-r-1} = \frac{1}{5} \hat{\varepsilon}^T \hat{\varepsilon} \sim 2$

Block Hypothesis

$$\beta = \begin{bmatrix} \beta_0 \\ \vdots \\ \beta_r \end{bmatrix} = \begin{bmatrix} \tilde{\beta}_1 \\ \tilde{\beta}_2 \end{bmatrix} \rightarrow \begin{bmatrix} \beta_0 \\ \vdots \\ \beta_q \end{bmatrix}_{(q+1) \times 1} \begin{bmatrix} \beta_{q+1} \\ \vdots \\ \beta_r \end{bmatrix}_{(r-q) \times 1}$$

$$H_0 : \beta_{q_{t+1}} = \beta_{q_{t+2}} = \dots = \beta_z = 0$$

H_1 : at least one of them non zero

i.e.

$$H_0 : \tilde{\beta}_z = 0$$

$$H_1 : \tilde{\beta}_z \neq 0$$

used in
feature
reduction.

$$z = \begin{bmatrix} \tilde{z}_1 & | & \tilde{z}_2 \end{bmatrix}$$

order needs to
be preserved in
 z block also

$$y = \tilde{z}_1 \tilde{\beta}_1 + \tilde{z}_2 \tilde{\beta}_2 + \varepsilon$$

wilk's
lambda

$$\Lambda = \frac{\left(\hat{\sigma}_{MLE}^2 \right)_{\tilde{z}_1}}{\left(\hat{\sigma}_{MLE}^2 \right)_z} \quad \begin{matrix} -n/2 \\ \tilde{z}_1 \text{ block} \\ \text{complete } z \text{ block} \end{matrix}$$

$$= \left(1 + \frac{\hat{\sigma}_{MLE, z_1}^2 - \hat{\sigma}_{MLE, z}^2}{\hat{\sigma}_{MLE, z}^2} \right)^{-n/2}$$

reject H_0 : $\lambda < c$

↑ ↙

small for critical value

rejection

$$\frac{\hat{\sigma}_{MLE, z_1}^2 - \hat{\sigma}_{MLE, z}^2}{\hat{\sigma}_{MLE, z}^2} > \hat{c} \quad \text{reject } H_0$$

$$\frac{(y - \tilde{z}_1 \hat{\beta}_1)^T (y - \tilde{z}_1 \hat{\beta}_1) - (y - z \hat{\beta})^T (y - z \hat{\beta})}{(y - z \hat{\beta})^T (y - z \hat{\beta})}$$

$$= \frac{[ESS(z_1) - ESS(z)] / (r - q)}{ESS(z) / (n - r - 1)} \sim F_{r-q, n-r-1}$$